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Isomeric Yield Ratios of the (γ, *n*) and (*n*, 2*n*) Reactions on Nuclei of ^{110}Pd, ^{142}Nd, and ^{144}Sm

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Abstract—The isomeric yield ratios of the (γ, *n*) and (*n*,2*n*) reactions on nuclei of ^{110}Pd, ^{142}Nd, and ^{144}Sm are measured by the induced radioactivity method. The energy dependence of the isomeric yield ratios of the photonuclear reactions ^{110}Pd(γ, *n*)^{109m,g}Pd, and ^{142}Nd(γ, *n*)^{141m,g}Nd, and ^{144}Sm(γ, *n*)^{143m,g}Sm is investigated in the energy range 14–35 MeV. The results are compared to other published experimental data.

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INTRODUCTION

Study of the isomeric yield ratios and cross-sections of photonuclear reactions allows us to obtain information on their spin distribution levels and nuclear level density parameters, the parameters of the statistical model's spin restriction, and the contribution from direct processes to the mechanism of photonuclear reactions [1, 2]. Data on the isomeric yield ratios of photonuclear reactions are necessary for γ activation analysis [3], without which it is impossible to assess the method's theoretical possibilities, or to develop optimal techniques.

In this work, the energy dependence of the isomeric yield ratio $d(E_{\gamma max}) = Y_m/Y_g$ of the photonuclear reaction of the (γ,*n*) type on nuclei of ^{110}Pd, ^{142}Nd, and ^{144}Sm is investigated by the induced radioactivity method. We also study the isomeric yield ratio in the (n,2n) reaction on these nuclei.

Our studies were conducted using the SB-50 high-current betatron at the Research Institute of Applied Physics, National University of Uzbekistan, and using the NG-150 neutron generator at the Institute of Nuclear Physics, Republic of Uzbekistan Academy of Sciences.

The (γ, *n*) reaction experiments were conducted on the braked γ beam of the SB-50 betatron in the energy range 10–35 MeV with a step of 1 MeV. The time regimes (i.e., the time of irradiation, pauses, and measurements) were chosen in accordance with the half-life of the formed radioactive nuclides. As targets, we used palladium, neodymium, and samarium in a natural mixture of isotopes. To increase the dose power, we performed the irradiation inside the acceleration chamber of the SB-50 high-current betatron at a distance of 12 cm from the tungsten braking target where the sample inside the special container was delivered with the help of a type K5-2A pneumatic transport setup. It took the transport setup ~4 s [4] to deliver the sample to the irradiation site.

The neutron source was an NG-150 neutron generator that generated fluxes of fast neutrons with energies of ~2.4 and 14 MeV from reactions of d + d → ^{3}He + n or t + d → α + n using deuterium and tritium targets. The neutron fluxes were about ~10^8 and 10^{10} neutron s^{-1}, respectively [5]. The time of irradiation by the neutron flux at an energy of 14 MeV was 10 min.

The induced activity of targets was measured using a Canberra γ spectrometer consisting of an HPGe germanium detector with a relative efficiency of 15% and a resolution for the 1332-keV Co line of 1.8 keV, a DSA 1000 digital analyzer, and a personal computer with the Genie 2000 software package for the acquisition and processing of γ spectra. The gamma spectrometer was energy calibrated using the OSGI standard set of sources. Measurements were made in the standard geometry in which the detector was calibrated with respect to efficiency. Measurements of the gamma spectra of targets began after a pause of 0.5–50 min and proceeded for 3–120 min.

The filling of the isomeric and ground levels was identified according to their γ lines. The spectroscopic characteristics of the product nuclei of the (γ, *n*) and (n, 2n) reactions, required for processing the measurement results, are taken from [6] and given in Table 1, where I^π is the spin and parity of the level, $T_{1/2}$ is the nucleus half-life, I_γ is the intensity of γ quanta of the

Table 1. Spectroscopic characteristics of the product nuclei of (γ, *n*) and (*n*, 2*n*) reactions

Product nucleus	I^{π}	$T_{1/2}$	E_{γ}, keV	I_{γ}, %	*p*
^{95m}Pd	$11/2^{-}$	4.69 min	188.9	58.00	1
^{95g}Pd	$5/2^{+}$	13.47 h	88.10	5.00	–
^{141m}Nd	$11/2^{-}$	60.3 s	756.5	91	0.9997
^{141g}Nd	$3/2^{+}$	2.6 h	91.10 120.50 275.40 319.40 398.20 439.80 531.00 685.90	28.30 0.39 1.00 2.19 0.90 1.19 13.50 0.79	–
^{143m}Sm	$11/2^{-}$	65 s	690.00 754.4 963.00	1.67 100 1.13	0.998
^{143g}Sm	$1/2^{+}$	8.83 min	511 1056.6	100 2.6	–

fixed decay energy, and *p* is the branching coefficient of the γ transition.

Isomeric yield ratios were calculated according to the formula [7]

$$d = \frac{Y_{\mathrm{m}}}{Y_{\mathrm{g}}} = \left[\frac{\lambda_{\mathrm{g}}F_{\mathrm{m}}(t)}{\lambda_{\mathrm{m}}F_{\mathrm{g}}(t)}\left(C\frac{N_{\mathrm{g}}I_{\mathrm{m}}\varepsilon_{\mathrm{m}}}{N_{\mathrm{m}}I_{\mathrm{g}}\varepsilon_{\mathrm{g}}} - p\frac{\lambda_{\mathrm{g}}}{\lambda_{\mathrm{g}}-\lambda_{\mathrm{m}}}\right) + p\frac{\lambda_{\mathrm{m}}}{\lambda_{\mathrm{g}}-\lambda_{\mathrm{m}}}\right]^{-1}, \quad (1)$$

where

$$F_{\mathrm{m}}(t) = \left[1-\exp(-\lambda_{\mathrm{m}}t_{\mathrm{o}})\right]\exp(-\lambda_{\mathrm{m}}t_{\mathrm{n}})\left[1-\exp(-\lambda_{\mathrm{m}}t_{\mathrm{c}})\right], \quad (2)$$

$$F_{\mathrm{g}}(t) = \left[1-\exp(-\lambda_{\mathrm{g}}t_{\mathrm{o}})\right]\exp(-\lambda_{\mathrm{g}}t_{\mathrm{n}})\left[1-\exp(-\lambda_{\mathrm{g}}t_{\mathrm{c}})\right], \quad (3)$$

λ_{m} and λ_{g} are the decay constants of the isomeric and ground states, respectively; *C* is a coefficient that allows for miscounts of the recording apparatus and the positions of pulses; N_{m} and N_{g} are the numbers of decay events for the isomeric and ground states respectively; *I* is the intensity of γ quanta with the given decay energy; ε is the spectrometer efficiency; *p* is the branching coefficient of the γ transition; and t_0, t_{n}, and t_{c} are the irradiation time, the time of pause, and the measurement time respectively.

The experimental results obtained with regard to the isomeric yield ratios of the (γ, *n*) and (*n*, 2*n*) reactions on the ^{110}Pd, ^{142}Nd, and ^{144}Sm nuclei are given in the figure and in Tables 2 and 3. The absolute error of isomeric yield ratios is determined by the statistical error of counts in the photopeak of the measured γ line and by the efficiency of the γ radiation recording. The measurement results showed that values of the isomeric yield ratio $Y_{\mathrm{m}}/Y_{\mathrm{g}}$ rose from the reaction's threshold to the energy corresponding to the position of the maximum of the giant dipole resonance, E_{m}, which was 15.92, 14.94, and 15.32 MeV for the photoneutral reactions on nuclei of ^{110}Pd, ^{142}Nd, and ^{144}Sm, respectively [8]. This could be because the number of cascade γ transitions relaxing the nucleus excitation grows as the energy and the number of moments carried away by quasidirect neutrons increases. The function $d(E_{\gamma\max}) = Y_{\mathrm{m}}/Y_{\mathrm{g}}$ for energies higher than E_{m} has the form of saturation curve.

The experimental results of our measurements are given in Tables 2 and 3 for comparison with data obtained previously. As we can see, there is good agreement among the results of all works with regard to the isomeric ratios for the ^{142}Nd(γ, *n*) ^{141m,g}Nd reaction in the energy range 15–25 MeV within the limits of measurement errors (see Table 2), except for the

Table 2. Isomeric yield ratios of (γ, *n*)-type reactions on nuclei of ^{110}Pd, ^{142}Nd, and ^{144}Sm

	$E_{\gamma max}$, MeV	Y_m/Y_g	Reference
^{110}Pd(γ, *n*)^{109m,g}Pd	15.0	0.060 ± 0.005	[9]
	16.5	0.062 ± 0.005	[9]
	18.5	0.068 ± 0.005	[9]
	20.5	0.072 ± 0.006	[9]
	25	0.060 ± 0.007**	[10]
	25	0.065 ± 0.003	This work
	30	0.063 ± 0.003	This work
	55	0.11 ± 0.02	[11]
^{142}Nd(γ, *n*)^{141m,g}Nd	15.0	0.022 ± 0.002	[9]
	16.5	0.045 ± 0.004	[9]
	16	0.056 ± 0.010*	[2]
	18	0.049 ± 0.004	[9]
	20.5	0.052 ± 0.006	[9]
	18	0.046 ± 0.003	[12]
	18	0.043 ± 0.002	This work
	20	0.045 ± 0.002	This work
	25	0.043 ± 0.002	[13]
	25	0.061 ± 0.010**	[10]
	30	0.045 ± 0.001	[13]
	55	0.055 ± 0.0006	[11]
	70	0.19 ± 0.01	[14]
^{144}Sm(γ, *n*)^{143m,g}Sm	15.0	0.031 ± 0.003	[9]
	16.5	0.039 ± 0.003	[9]
	16	0.057 ± 0.010*	[2]
	18.0	0.043 ± 0.003	[9]
	20.5	0.044 ± 0.003	[9]
	25	0.047 ± 0.004**	[10]
	25	0.041 ± 0.002	This work

Notes: * The given value is $r = \sigma^m/\sigma^g$.

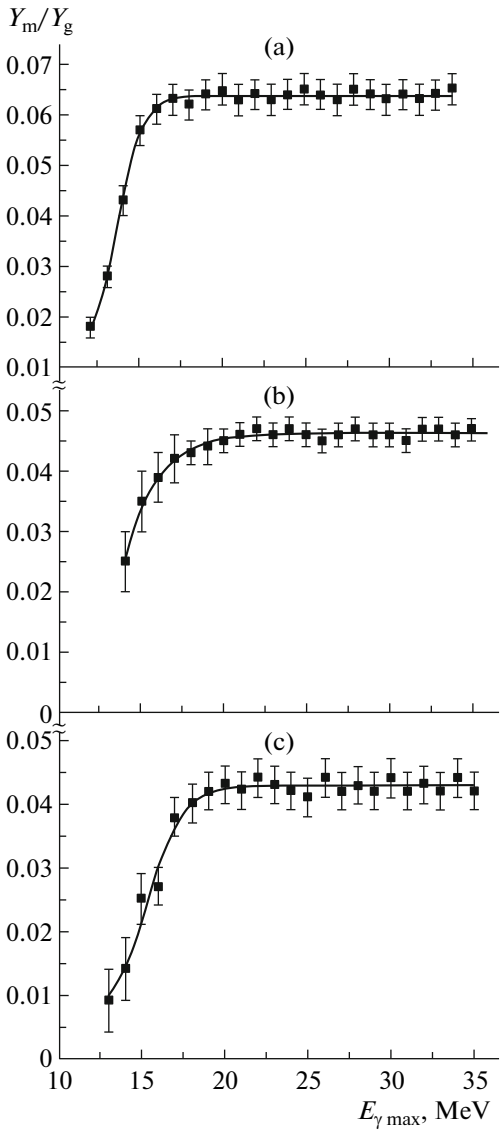
** The given value is $r_{int} = \sigma^m_{int}/\sigma^g_{int}$.

Table 3. Isomeric yield ratios of (*n*, 2*n*)-type reactions on nuclei of ^{110}Pd, ^{142}Nd, and ^{144}Sm

Reaction	Y_m/Y_g	Reference
^{110}Pd(*n*, 2*n*)^{109m,g}Pd	0.41 ± 0.03	[9]
	0.449 ± 0.049	[15]
	0.360 ± 0.023	[16]
	0.42 ± 0.03	This work
^{142}Nd(*n*, 2*n*)^{141m,g}Nd	0.46 ± 0.04	[9]
	0.463 ± 0.051	[15]
	0.537 ± 0.061	[17]
	0.47 ± 0.04	This work
^{144}Sm(γ, *n*)^{143m,g}Sm	0.70 ± 0.06	[9]
	0.871 ± 0.096	[15]
	0.635 ± 0.088	[17]
	0.782 ± 0.078	[18]
	0.75 ± 0.07	This work

data in [10, 11, 14]. The energy dependence of isomeric yield ratios of the ^{142}Nd(γ, *n*) ^{141m,g}Nd was determined for the first time in the range of excitation energies higher than giant dipole resonance (i.e., in the 21–35 MeV range). The values of the isomeric ratios in the saturation region of the energy dependence curve for the isomeric yield ratios obtained in [11] are somewhat higher than those of other works (see Table 2). This could be because they were obtained in [11] at a substantially higher energy (E_γ = 55 MeV). The results of [14] differ sharply from the others. We cannot exclude the possibility that this is associated with the particulars of the energy dependence of isomeric ratios.

The results that we obtained for the ^{110}Pd(γ, *n*) ^{109m,g}Pd reaction within the error limits agree with the data of all other works except [11]. The isomeric ratios for the



Energy dependence of the isomeric yield ratios of reactions (a) ^{110}Pd(γ, *n*) ^{109m,g}Pd, (b) ^{142}Nd(γ, *n*) ^{141m,g}Nd, and (c) ^{144}Sm(γ, *n*) ^{143m,g}Sm.

^{144}Sm(γ, *n*) ^{143m,g}Sm reaction agree within the measurement errors. As can be seen from Table 2, the isomeric ratios of the cross sections [2] and those of the integral cross sections [11] are slightly higher for the ^{142}Nd(γ, *n*) ^{141m,g}Nd and ^{144}Sm(γ, *n*) ^{143m,g}Sm reactions than the isomeric ratios of the yields. This can be explained by the discrepancy between the thresholds of the (γ, *n*) and (γ, *n*)^m reactions.

In the case of reaction (n,2n) (see Table 3), the data of all the works are in good agreement within the measurement errors. The isomeric ratio values obtained in [8] are somewhat underestimated with respect to the results of other works.

CONCLUSIONS

From our analysis of the data given in Tables 2 and 3, it follows that the experimental investigations on the exciting of isomeric states in photonuclear reactions of the (γ, *n*) type were mainly conducted in the energy range 10–25 MeV, i.e., in the region of giant dipole resonance. In the region of energies higher than the giant dipole resonance, the energy dependence of isomeric ratios is not yet adequately studied. Thanks to these studies, we can obtain information on the nuclear level density and the contribution from direct processes to the mechanism of photonuclear reactions in the given range of energies.

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